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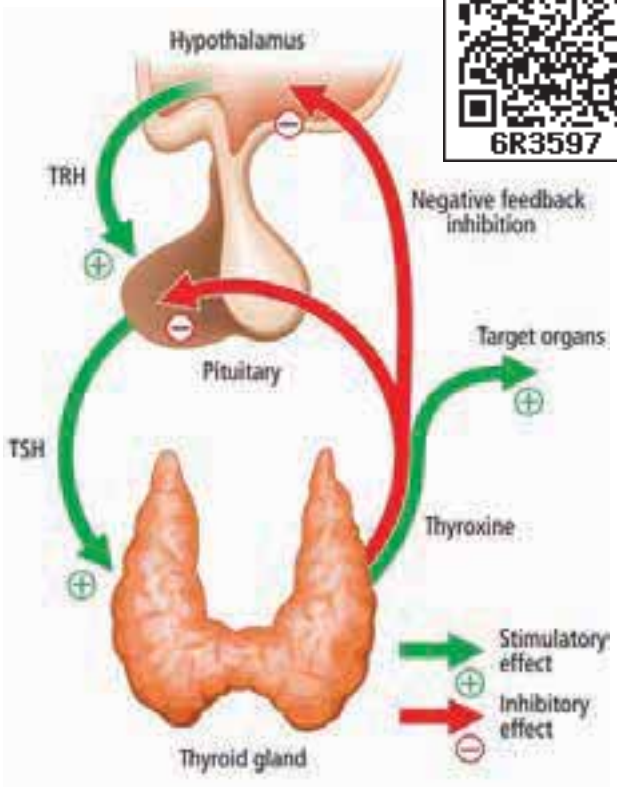

Figure: 11.4. Negative feedback mechanism

lumen of acinus is filled with colloid, a thick glycoprotein mixture consisting of thyroglobulin molecules.

Hormones of the thyroid gland are often called the major metabolic hormones. The follicular cells of thyroid gland secrete two hormones namely tri-iodothyronine (T_3) and thyroxine or tetra- iodothyronine (T_4). The parafollicular cells or 'C' cells of thyroid gland secrete a hormone called thyrocalcitonin. Iodine is essential for the normal synthesis of thyroid hormones. Thyroid releasing hormone from the hypothalamus stimulates the adenohypophysis to secrete TSH, which inturn stimulates the thyroid gland to secrete the thyroid hormones. Thyroid hormones show a negative feedback effect on the hypothalamus and pituitary (Figure11. 4).

Sporodic goitre is a genetic disease and is not caused by iodine or thyroxine deficiency.

Iodine is required for formation of thyroxine: To produce normal quantities of thyroxine, about 1mg/week of iodine is required. To prevent iodine deficiency common table salt is iodised with 1 part sodium iodide to every 1,00,000 parts of sodium chloride

Functions of thyroxine or tetra-iodothyronine (T4): Thyroxine regulates the basal metabolic rate (BMR) and body heat production. It stimulates protein synthesis and promotes growth. It is essential for the development of skeletal and nervous system. Thyroxine plays an important role in maintaining blood pressure. It reduces serum cholesterol levels, Optimum levels of thyroxine in blood is necessary for gonadial functions.

Functions of thyrocalcitonin (TCT): TCT is a polypeptide hormone, which regulates the blood calcium and phosphate levels. It reduces the blood calcium level and opposes the effects of parathyroid hormone.

11.2.5 Parathyroid gland

In human, four tiny parathyroid glands are found in the posterior wall of the thyroid glands. This gland is composed of two types of cells, the chief cells and oxyphil cells. The chief cells secrete parathyroid hormone (PTH) and the functions of oxyphil cells are not known.